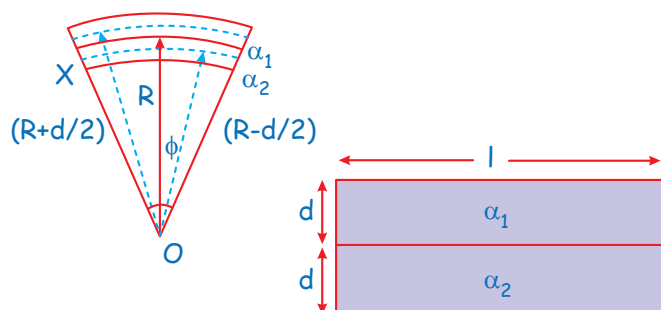


A bimetallic strip (b) when heated and (c) when cooled. Often brass [$\alpha = 19 \times 10^{-6} \text{ }^\circ\text{C}^{-1}$] and steel [$\alpha = 12 \times 10^{-6} \text{ }^\circ\text{C}^{-1}$] are selected. The two pieces are welded or riveted together. When the bimetallic strip is heated, the brass having the larger value of α expands more than the steel. Since, the two metals are bonded together, the bimetallic strips bends into an arc as in part b, with the longer brass piece having a larger radius than the steel piece. When the strip is cooled, the bimetallic strips bends in the opposite direction, as in part c.

Mathematical Solution



$$\left(R + \frac{d}{2}\right) \phi = L_0 (1 + \alpha_1 \Delta\theta)$$

$$\left(R - \frac{d}{2}\right) \phi = L_0 (1 + \alpha_2 \Delta\theta)$$

$$\frac{R + \frac{d}{2}}{R - \frac{d}{2}} = \frac{1 + \alpha_1 \Delta\theta}{1 + \alpha_2 \Delta\theta}$$

From the above equation we can find the value of R.

TEMPERATURE SCALES

For any types of scale

$$\frac{\text{Reading point} - \text{ice point}}{\text{Steam point} - \text{ice point}} = \text{constant}$$

The relation between these scales is given by the following equation.

$$\frac{C - 0^\circ}{100^\circ - 0^\circ} = \frac{K - 273}{373 - 273} = \frac{F - 32}{212 - 32} \Rightarrow \frac{C}{5} = \frac{K - 273}{5} = \frac{F - 32}{9}$$

C = Reading in Celsius

K = Reading in Kelvin

F = Reading in Fahrenheit

CALORIMETRY

Heat

When a hot body is put in contact with a cold one, the former gets colder and the latter warmer. From this observation it is natural to conclude that a certain quantity of heat has passed from the hot body to the cold one. Heat is a form of energy.

$$\begin{array}{ccc} \text{m} & \xrightarrow{\Delta Q = ?} & \text{m} \\ T_1 & & T_2 \\ \Delta Q \propto m & & \\ \alpha \Delta T & & \\ \hline \Delta Q = m S \Delta T \end{array}$$

Here ΔQ = Heat given or taken

m = mass of the body

S = Specific heat constant or specific heat

ΔT = Change in temp

mS = Heat capacity

$$\star \Delta Q = m S \Delta T$$

$$\star dQ = m S dT$$

$$\star \text{In differential form } S = \frac{1}{m} \frac{dQ}{dT}$$

★ S.I. unit of s is joule/kg-kelvin

★ **Specific heat of water:** $s = 1 \text{ cal/gm}^\circ\text{C}$

★ **Specific heat of ice:** $s = 0.5 \text{ cal/gm}^\circ\text{C}$

★ **Specific heat of steam:** $s = 0.46 \text{ cal/gm}^\circ\text{C}$
 $\approx 0.5 \text{ cal/g}^\circ\text{C}$

Q. Find the amount of heat required to raise the temperature of 50 gm water at 20°C to 60°C .
Given $S_{\text{water}} = 1 \text{ cal/gm}^\circ\text{C}$

Sol. 50 gm water $20^\circ\text{C} \xrightarrow{\Delta Q = ?} 50 \text{ gm water } 60^\circ\text{C}$
 $\Delta Q = m S \Delta T = 50 \text{ gm} \cdot \frac{1 \text{ cal}}{\text{gm}^\circ\text{C}} \times (60 - 20)^\circ\text{C} = 200 \text{ cal}$

Q. Find amount of heat release from 50 gm water at 50°C to 50 gm water 20°C .

Sol. $\Delta Q = m S \Delta T$ (magnitude) = $50 \times 1 \times 30 = 1500 \text{ cal}$

Q. 50 gm ice $-20^\circ \xrightarrow{\Delta Q = ?} 50 \text{ gm water } -5^\circ\text{C}$

Sol. $\Delta Q = m S_{\text{ice}} \Delta T = 50 \times \frac{1}{2} \times 15 = 375 \text{ cal}$

Specific Heat Constant (S or C)

It is the amount of energy required to raise the temperature of unit mass of that substance by 1°C (or 1 K) is called specific heat. It is represented by S or C

$$\Delta Q = mS\Delta T \quad d\theta = mSdT$$
$$S = \frac{\Delta Q}{m\Delta T} \quad S = \frac{1}{m} \frac{dQ}{dT}$$

- ★ Thermal capacity = mass $\times$ specific heat = ms
- ★ 1 calorie is defined as amount of heat required to raised the temperature of 1 gm of water from 14.5°C to 15.5°C .
- ★ $1\text{ cal} = 4.182\text{ J} = 4.2\text{ J}$

Molar Heat Capacity

The amount of energy needed to raise the temperature of one mole of a substance by 1°C (or 1 K) is called molar heat capacity.

Molar specific heat capacity, $C = \frac{1}{n} \left(\frac{dQ}{dT} \right)$ where n is no. of moles

Q. The specific heat of a substance is given by

$C = a + bT$, where $a = 1.12\text{ kJ kg}^{-1}\text{C}^{-1}$ and $b = 0.016\text{ kJ kg}^{-1}\text{C}^{-1}$. The amount of heat required to raise the temperature of 1.2 kg of the material from 280 K to 312 K is:

Sol. Heat required

$$= Q = \int_{280}^{312} mCdT = \int_{280}^{312} (1.2)(a + bT)dT$$
$$= 1.2 \left[aT + \frac{bT^2}{2} \right]_{280}^{312}$$
$$= 1.2 [1.12 (312 - 280) + \frac{0.016}{2} \{(312)^2 - (280)^2\}]$$
$$= 1.2 [1.12 \times 32 + \frac{0.016}{2} \times (97344 - 78400)]$$
$$= 1.2 [35.84 + 0.008 \times 18944]$$
$$= 224.8704\text{ KJ} = 225\text{KJ}$$

Mechanical Equivalent of Heat

According to Joule, work may be converted into heat and vice-versa. The ratio of work done to heat produced is always constant. $\frac{W}{H} = \text{constant (J)}$
 $\Rightarrow W = JH$

J is called mechanical equivalent of heat.

$J = 4.18\text{ joule/cal}$

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WATER EQUIVALENT

- ★ Amount of water that absorbs or gives out the same amount of heat as it done by the body when heated or cooled through 1°C .
- ★ Mass of water having the same thermal capacity as the body is called water equivalent of body.

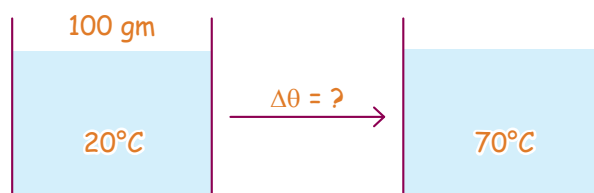
$$\Delta Q = m_b S_b \Delta T = m_w S_w \Delta T \quad (S_w = 1)$$

$$m_w = \frac{m_b S_b}{S_w} = \text{water equivalent} = m_b S_b \text{ (value)}$$

Q. Find the amount of heat required to raise the temp of 100 g water in following figure by 50° .

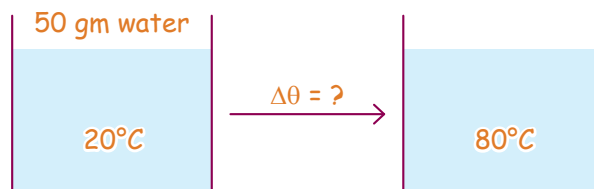
$m_b = 200\text{ gm} = \text{mass of beaker}$

$S_b = 10\text{ cal/gm}^{\circ}\text{C} = \text{specific heat of beaker}$



Sol. $\Delta\theta = m_w S_w \Delta T + m_b S_b \Delta T$
 $= 100 \times 1 \times 50 + 200 \times 10 \times 50 = 105000\text{ cal}$

Q. Find the amount of heat required to raise the temp of 50 gm water by 60°C in following ques. If water equivalent of beaker is 30 gm.



Sol. $\Delta\theta = m_w S_w \Delta T + m_b S_b \Delta T$
 $\Delta\theta = 50 \times 1 \times 60 + 30 \times 60$
 $\Delta\theta = (50 + 30) \times 1 \times 60 = 4800$

Phase Change

- ★ If we supply heat, apart from rising the temperature, it may change the state of body/substance.

Example: Solid $\longrightarrow$ Liquid

Liquid $\longrightarrow$ Vapour

★ If m mass of substance changes its state then

$$\Delta\theta \propto m$$

$$\Delta\theta = mL$$

→ Latent heat

Latent Heat or Hidden Heat

When state of a body changes, change of state takes place at constant temperature [melting point or boiling point] and heat released or absorbed is $Q = mL$ where L is latent heat. Heat is absorbed if solid converts into liquid (at melting point) or liquid converts into vapours (at boiling point) and heat is released if liquid converts into solid or vapours converts into liquid.

★ **Latent heat of fusion:** It is the quantity of heat (in kilocalories) required to change its 1 kg mass from solid to liquid state at its melting point.

★ Latent heat of fusion for ice: 80 cal/g.

★ **Latent heat of vaporization:** The quantity of heat required to change its 1 kg mass from liquid to vapour state at its boiling point.

★ Latent heat of vaporisation for water:
= 536 cal/g $\cong$ 540 cal/g (approx)

चलो भाई बहुत हो गई definition-definition अब ques की practice करते हैं। BCZ we have no. of page contrained अब मैं ques में short language use करूंगा..... वरना publisher वाले sir book की price बढ़ा देंगे।



Q. Found the amount of heat required to melt 20 g of ice at 0°C .

20 gm ice 0° $\xrightarrow{\Delta\theta = ?}$ 20 gm water 0°C
(solid) (liquid)

Sol. $\Delta Q = mL = 20 \times 80 = 1600 \text{ cal}$

Q. 10 gm water $\xrightarrow{\Delta Q}$ 10 gm vapor
100°C 100°C

Sol. $\Delta Q = mL_v = 10 \times 540 = 5400 \text{ cal}$

Q. 10 gm water $\xrightarrow{\Delta Q = ?}$ 8 gm water 2 gm vapour
100°C 100°C

Sol. $\Delta\theta = mL_v = 2 \times 540 = 1080 \text{ cal}$

Q. Find the amount of heat required to change 10 gm water at 20°C to 6 gm water and 4 gm vapour at 100°C .

10 gm water $\xrightarrow{\Delta Q = ?}$ 6 gm water (at 100°C)
20°C 4 gm vapour

Sol.

सबसे पहले water का temp 20°C से $m\Delta T$ लगाकर 100°C water तक ले जाओ, फिर $\Delta Q = mL$ लगाकर 4 gm water का phase change करदो।



10 gm water $\xrightarrow{\Delta Q = ?}$ 6 gm water (at 100°C)
 20°C
 $\Delta\theta_1 = mS\Delta T$
 10 gm water 100°C
 4 gm vapour
 $\Delta\theta_2 = m'L_v$

$$\Delta\theta = mS\Delta T + m'L_v$$

$$\Delta\theta = 10 \times 1 \times 80 + 4 \times 540$$

Q. Find the amount of heat required to change 10 gm ice at -20°C to 6 gm water + 4 gm ice at 0°C .

Sol.

10 gm ice $\xrightarrow{\Delta\theta_{\text{net}}}$ 4 gm ice
 -20°C 6 gm Water at 0°C

$\Delta\theta_1 = mS_{\text{ice}}\Delta T$ 100°C $\Delta\theta_2 = m'L_f$

10 gm ice 0°C

$$\Delta\theta_{\text{net}} = 10 \times \frac{1}{2} \times 20 + 6 \times 80$$

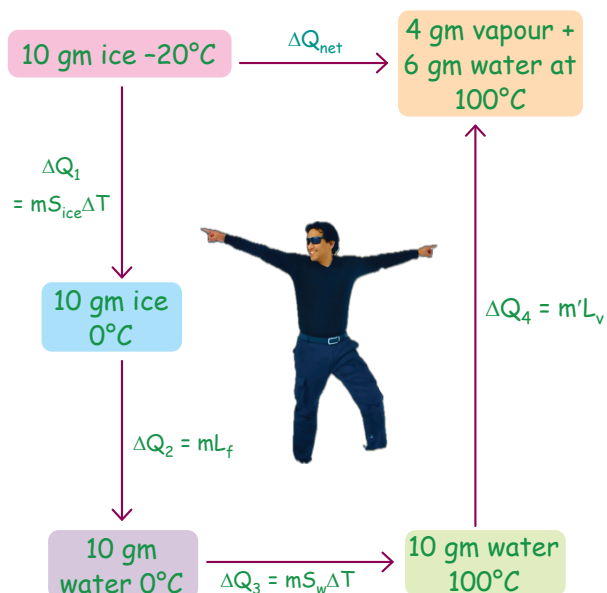
Q. 10 gm ice $\xrightarrow{\Delta Q_{\text{net}} = \Delta Q_1 + \Delta Q_2 + \Delta Q_3}$ 10 gm water 50°
 -20°C

Sol. $\Delta Q_{\text{net}} = mS_{\text{ice}}\Delta T_1 + mL_f + mS_w\Delta T_2$

$$= 10 \times \frac{1}{2} \times 20 + 10 \times 80 + 10 \times 1 \times 50$$

Q. Find amount of heat required to change 10 gm ice at -20°C to 4 gm vapour + 6 gm water at 100°C .

Sol.



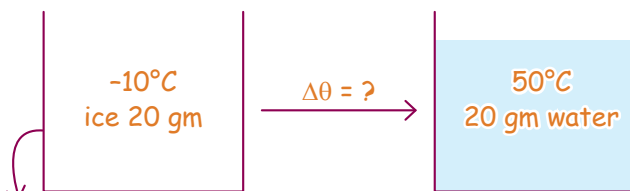
$$\Delta Q_{\text{net}} = 10 \times \frac{1}{2} \times 20 + 10 \times 80 + 10 \times 1 \times 100 + 4 \times 540$$

= जोड़ के खुद लिख ले ये भी मैं ही करू क्या

Q. 20 gm ice -40°C $\longrightarrow$ 20 gm vapour 100°C

Sol. $\Delta\theta = 20 \times \frac{1}{2} \times 40 + 20 \times 80 + 20 \times 1 \times 100 + 20 \times 540$

Q.



Water equilateral = 10 gm

Sol. $\Delta\theta = \left(20 \times \frac{1}{2} \times 10 + 20 \times 80 + 20 \times 1 \times 50 \right) + 10 \times 60$

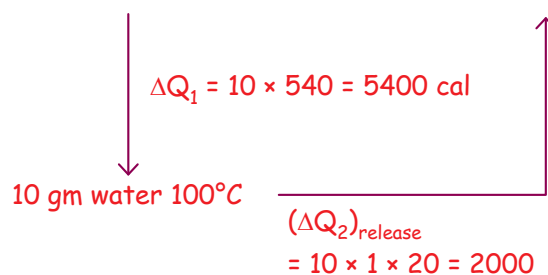
Q. 40 gm water 50° $\xrightarrow{\Delta Q_{\text{release}} = 40 \times 1 \times 40 = 1600}$ 10 gm water 10°C

Q. 10 gm water 10°C $\longrightarrow$ 10 gm ice 0°C

Sol. Amount of heat release

$$= 10 \times 80 = 800$$

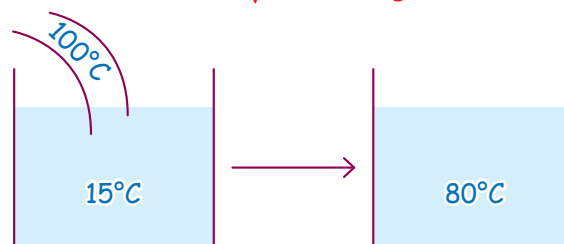
Q. 10 gm Steam 100°C $\xrightarrow{\Delta\theta_{\text{release}} = ?}$ 10 gm water 80°C



Sol. Amount of heat release = $10 \times 540 + 10 \times 1 \times 20$

Q. Steam at 100°C is passed into 1.1 kg of water contained in a calorimeter of water equivalent .02 kg at 15°C till the temperature of calorimeter and its contents rises to 80°C . What is mass of steam condensed. $L_v = 540 \text{ cal/gm}$.

Sol.



$$m_w S_w (80 - 15) + m_b S_b (80 - 50)$$

$$= m_{\text{steam}} \times 540 + m_{\text{steam}} S_w \times 20$$

$$1.12 \times 10^3 \times 1 \times 65 = m \times 540 + m \times 1 \times 20$$

Q. 500 gm of water at 80°C is mixed with 100 gm steam at 120°C . Find out the final mixture.

Sol. 120°C steam $\longrightarrow$ 100°C steam

$$\text{heat released} = 100 \times 0.46 \times 20 = 920 \text{ cal}$$

80°C water $\longrightarrow$ 100°C water

$$\text{Req. heat} = 500 \times 1 \times 20 = 10 \text{ kcal}$$

100gm steam $\longrightarrow$ 100 gm water at 100°C

$$\text{heat released} = 100 \times 540 = 54 \text{ kcal}$$

$$\text{Total heat released} = 55 \text{ kcal.}$$

$$\text{Remaining heat} = 55 - 10 = 45 \text{ kcal}$$

Now we have 600 gm water at 100°C

$$\Rightarrow 45000 = m \times 540 \Rightarrow m = \frac{250}{3} \text{ gm}$$

So at last we have $\frac{250}{3}$ gm steam and

$$\left(600 - \frac{250}{3} \right) \text{ gm of water}$$

Q. Find amount of heat required if 100 g ice at -10°C is converted into 100 g steam at 120°C .

Sol. -10°C , 100 gm ice $\longrightarrow$ 120° 100 gm steam

$$\begin{aligned}
 Q &= ms\Delta T \\
 &= \frac{1}{2} \times 100 \times 10 \\
 &= 500 \text{ cal.} \\
 0^{\circ}, 100 \text{ gm ice} \\
 Q &= mL_f \\
 &= 100 \times 80 \\
 &= 8000 \\
 0^{\circ}, 100 \text{ gm water} \\
 Q &= ms\Delta T \\
 &= 100 \times 1 \times 100 \\
 &= 10000 \text{ cal.} \\
 100^{\circ}, 100 \text{ gm water} \\
 Q &= mL_v \\
 &= 100 \times 540 \\
 &= 54000 \text{ cal} \\
 100^{\circ}, 100 \text{ gm Steam}
 \end{aligned}$$

$$\begin{aligned}
 Q &= ms\Delta T \\
 &= 100 \times 0.46 \times 20 \\
 &= 920 \text{ cal.}
 \end{aligned}$$

$$Q_{\text{net}} = 73.42 \text{ kcal.}$$

Q. The specific heat of a substance is given by

$C = a + bT$, where $a = 1.12 \text{ kJ kg}^{-1}\text{C}^{-1}$ and

$b = 0.016 \text{ kJ kg}^{-1}\text{C}^{-1}$. The amount of heat required to raise the temperature of 1.2 kg of the material from 280 K to 312 K is:

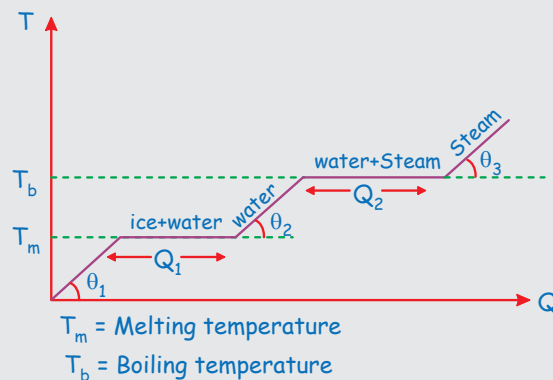
Sol. Heat required

$$\begin{aligned}
 Q &= \int_{280}^{312} mC dT = \int_{280}^{312} (1.2)(a + bT) dT \\
 &= 1.2 \left[aT + \frac{bT^2}{2} \right]_{280}^{312} \\
 &= 1.2 \left[1.12 (312 - 280) + \frac{0.016}{2} \{ (312)^2 - (280)^2 \} \right] \\
 &= 1.2 \left[1.12 \times 32 + \frac{0.016}{2} \times (97344 - 78400) \right] \\
 &= 1.2 [35.84 + 0.008 \times 18944] \\
 &= 224.8704 \text{ KJ} \\
 &= 225 \text{ KJ}
 \end{aligned}$$

Heating Curve

Heating Curve

If to a given mass (m) of a solid, heat is supplied at constant rate and a graph is plotted between temperature and time as shown in figure is called heating curve.



If heat is supplied at constant rate

$$dQ = msdT$$

$$\text{Slope} = \frac{dT}{dQ} = \frac{1}{ms}$$

$$\text{Since } s_{\text{ice}} = s_{\text{steam}} = 0.5 \text{ cal/g}^{\circ}\text{C}$$

$$\text{and } s_{\text{water}} = 1 \text{ cal/g}^{\circ}\text{C}$$

$$\Rightarrow \theta_1 = \theta_3 > \theta_2 \text{ and } Q_2 > Q_1$$

Q. Suppose we have m mass of ice at -40°C and we are supplying heat at constant rate P . Draw the temperature vs time graph

Sol. When temperature of ice is increasing.

$$d\theta = ms dT$$

$$\frac{d\theta}{dt} = ms \frac{dT}{dt} = P$$

$$\text{Slope} = \frac{dT}{dt} = \frac{P}{ms} = \frac{P}{ms_{\text{ice}}} = \text{Constant}$$

When temperature of water is increasing

$$d\theta = ms dT$$

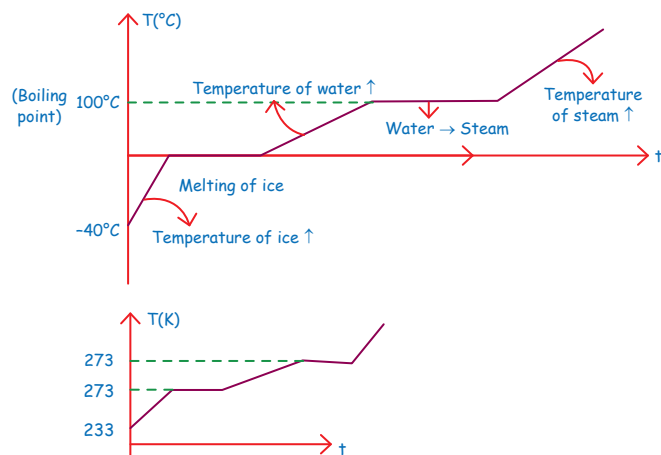
$$\frac{d\theta}{dt} = ms \frac{dT}{dt} = P = \text{Constant}$$

$$\text{Slope} = \frac{dT}{dt} = \frac{P}{ms} = \frac{P}{ms_{\text{water}}} = \text{Constant}$$

$$\therefore s_{\text{water}} > s_{\text{ice}}$$

$$(\text{Slope})_{\text{water}} < (\text{Slope})_{\text{ice}}$$

$$\frac{\tan \theta_2}{\tan \theta_1} = \frac{(\text{Slope})_{\text{water}}}{(\text{Slope})_{\text{ice}}} = \frac{S_i}{S_w} = \frac{1/2}{1} = \frac{1}{2}$$



Principle of Calorimetry

When two bodies at different temperature are allow to share heat, they attain a common temperature, and if assume that no heat is received from or given to any body outside the system and there is no chemical action involved in the process of sharing then heat given by one body

= heat loss by another body

Q. 40 gm water + 60 gm water $\longrightarrow$ T_f ?
 20°C 40°C of mixture
A B

Sol. $\Delta Q_1 + \Delta Q_2 = 0$ (from calorimetry principle)

$$40 \times 1 \times (T_f - 20) + 60 \times 1 \times (T_f - 40) = 0$$

$$4T_f - 80 + 6T_f - 240 = 0$$

$$T_f = 32^\circ\text{C}$$

Q. 40 gm + 60 gm + 40 gm $\longrightarrow$ T_f ?
 water water water of mixture.
 20°C 40°C 50°C

Sol. $\Delta\theta_1 + \Delta\theta_2 + \Delta\theta_3 = 0$ (from calorimetry principle)

$$40 \times 1 \times (T_f - 20) + 60 \times 1 \times (T_f - 40) + 40 \times 1 \times (T_f - 50) = 0$$

$$T_f = 37.1^\circ\text{C}$$

Q. A calorimeter of heat capacity 100 J/K is at room temperature of 30°C 100 gm of water at 40°C specific heat 4200 J/kg/K is poured into the calorimeter. What will be temperature of water in calorimeter.

Sol. $\Delta\theta_1 + \Delta\theta_2 = 0$

$$m_b S_b (T_f - 30) + m_w S_w (T_f - 40) = 0$$

$$100 (T_f - 30) + \frac{100}{1000} \times 4200 \times (T_f - 40) = 0$$

$$T_f = 38^\circ\text{C}$$

Q. 10 gm + 30 gm $\longrightarrow$ T_f ?
 $S_A = 20 \text{ cal/gm}^\circ\text{C}$ $S_B = 25 \text{ cal/gm}^\circ\text{C}$
 10°C 40°C

A

B

Sol. $\Delta\theta_1 + \Delta\theta_2 = 0$

$$10 \times 20 \times (T_f - 10) + 30 \times 25 \times (T_f - 40) = 0$$

$$T_f - 30 + \frac{32000}{950} \approx 33.68^\circ\text{C}$$

Q. In a container of negligible heat capacity, 200 gm ice at 0°C and 100 gm steam at 100°C are added to 200 gm of water that has temperature 55°C . Assume no heat is lost to the surroundings and the pressure in the container is constant 1.0 atm. (Latent heat of fusion of ice = 80 cal/gm, Latent heat of vaporization of water = 540 cal/gm, Specific heat capacity of ice = 0.5 cal/gm-K, Specific heat capacity of water = 1 cal/gm-K)

(a) What is the final temperature of the system?

(b) At the final temperature, mass of the total water present in the system, is

(c) Amount of the steam left in the system, is equal to

Sol. As steam has comparatively large amount of heat to provide in the form of latent heat we check what amount of heat is required by the water and ice to go up to 100°C , that is

$$(m_i L + m_i S_w \Delta T) + m_w S_w \cdot \Delta T$$

$$= [(200 \times 80) + (200 \times 1 \times 100)] + (200 \times 1 \times 45)$$

$$= 45,000 \text{ cal.}$$

That is given by m mass of steam, then

$$m_s L = 45,000$$

$$m_s = \frac{45,000}{540} = \frac{500}{6} = 83.3 \text{ gm}$$

Therefore 83.3 gm steam converts into water of 100°C .

$$\text{Total water} = 200 + 200 + 83.3 = 483.3 \text{ gm}$$

$$\text{Steam left} = 16.7 \text{ gm.}$$

HEAT TRANSFER

Mode of Heat Transfer

Heat is a form of energy which transfer from a body at higher temperature to a body at lower temperature. The transfer of heat from one body to another may take place by any one of the following modes:

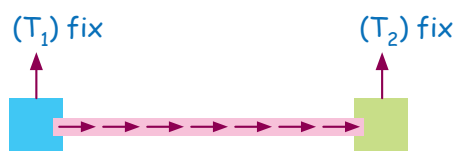
Conduction	Convection	Radiation
Heat Transfer due to Temperature difference	Heat transfer due to density difference	Heat transfer with out any medium
Due to free electron or vibration motion of molecules	Actual motion of particles	Electro-magnetic radiation
Heat transfer in solids body (in mercury also)	Heat transfer in fluids (Liquid + gas)	All
Slow process	Slow process	Fast process (3×10^8 m/sec)
Irregular path	Irregular path	Straight line (like light)

Conduction

The process by which heat is transferred from hot part to cold part of a body through the transfer of energy from one particle to another particle of the body without the actual movement of the particles from their equilibrium positions is called conduction.

Steady state: When temperature of the each cross-section of the bar becomes constant through different for different cross-sections is called steady state.

Suppose we have a rod whose temperature of one end is fixed at T_1 and another end is fix at T_2 . If $T_1 > T_2$ heat will start flowing from T_1 to T_2 .



After steady state we have observed rate of flow of heat $\frac{dQ}{dt}$

$$\frac{dQ}{dt} \propto \Delta T \quad (\text{Temp. difference})$$

$$\propto A \quad (\text{Cross - sectional Area})$$

$$\propto \frac{1}{l} \quad (l : \text{length of material})$$

$$\frac{dQ}{dt} = \frac{KA\Delta T}{l}$$

K = Coefficient of thermal conductivity

$$\frac{dQ}{dt} = \frac{KA\Delta T}{l}$$

(SI unit : W/mk
ie watt/meter kelvin)

★ In differential form this heat equation can be written as

$$\frac{dQ}{dt} = -KA \frac{dT}{dx}$$

Where $\frac{dT}{dx}$ is called temperature gradient

Conduction का ये इकलोता formula है जिस पर पूरा chapter टिका है, $\frac{dQ}{dt}$ मतलब rate of flow of heat,

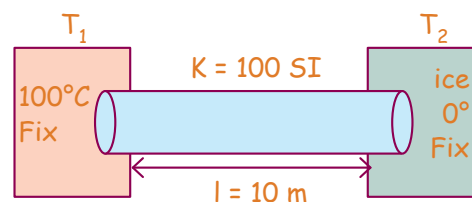
बोले तो per second कितनी heat transfer हो रही है

★ $\frac{dQ}{dt} = \frac{KA\Delta T}{l}$ (Magnitude)

★ $\frac{dQ}{dt} = -KA \frac{dT}{dx}$



Q. Find rate of flow heat and rate of melting of ice in following case.



Sol. Rate of flow of heat = $\frac{dQ}{dt} = \frac{KA\Delta T}{l}$

$$\frac{dQ}{dt} = \frac{100 \times 2 \times 100}{10} = 2000 \text{ J/sec}$$

$$\frac{dQ}{dt} = 2000 = \frac{d}{dt} (\text{ML})$$

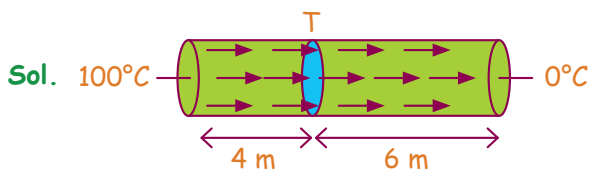
$$\Delta\theta = \text{mL}$$

$$\Rightarrow 2000 = L \frac{dm}{dt}$$

Now you can find $\frac{dm}{dt}$ which is rate of melting of ice but $L = 80$ put मत कर देना bcz यहाँ $\frac{dQ}{dt}$ joule per sec में है so, L की value ही joule/kg में put करना वरना पिटायी होगी।



Q. Find the temperature at a distance 4 m from hot end in above case.



At steady state $\frac{dQ}{dt}$ is same for all cross-section area.

$$\begin{aligned}\frac{dQ}{dt} &= \frac{KA(100 - 0)}{10} \\ &= \frac{KA(100 - T)}{4} = \frac{KA(T - 0)}{6}\end{aligned}$$

Now solve and get $\Rightarrow T = 60^\circ\text{C}$

SKC

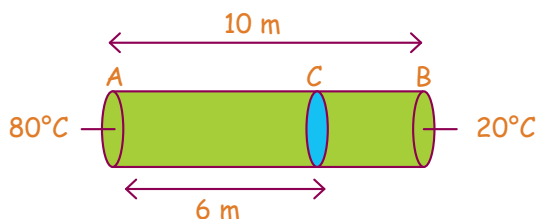
10 m में drop आया $\longrightarrow$ 100 का

1 m में drop आएगा $\longrightarrow \frac{100}{10} = 10$ का

So, 4 m में drop आएगा $\longrightarrow 10 \times 4 = 40$ का

Hence, 40°C का drop आया so, $T = 100 - 40 = 60^\circ\text{C}$

Q. Find temperature at C.



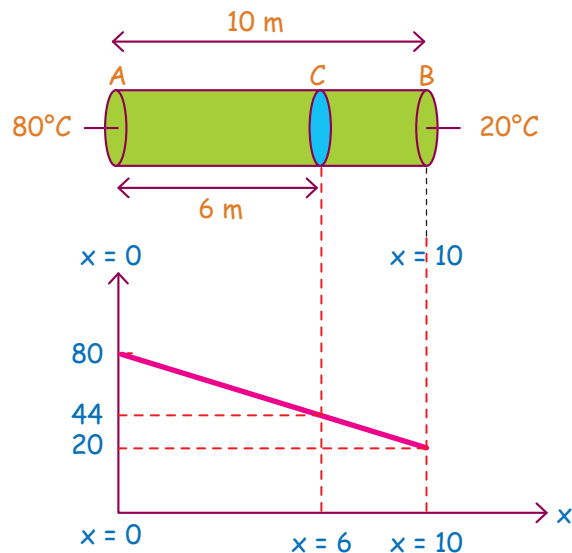
Sol. 10 m $\longrightarrow 60^\circ\text{C}$ (fall)
4 m $\longrightarrow \frac{60}{10} \times 4 = 24^\circ\text{C}$
 $T_B = 80 - 24 = 56$

$$\frac{dQ}{dt} = \frac{KA(80 - 20)}{10} = \frac{KA(80 - T)}{6} = \frac{KA(T - 20)}{4}$$

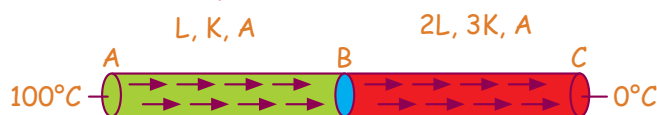
Solve any two eqn & get the answer

$T = 44$

Draw the graph between temperature and x if left end of rod at 80°C is at $x = 0$.



Q. Two different rod are connected to each other as shown in fig. Find temperature at junction B after steady state.



Sol. $\frac{dQ}{dt} = \frac{KA(100 - T)}{L} = \frac{3KA(T - 0)}{2L}$

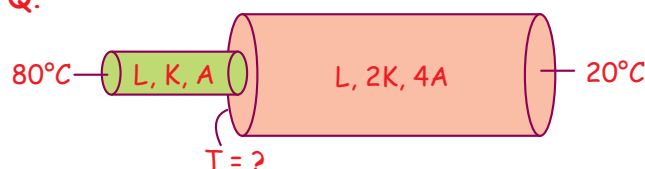
Same

$$200 - 2T = 3T$$

$$T = 40^\circ\text{C}$$

There is no accumulation of heat anywhere

Q.



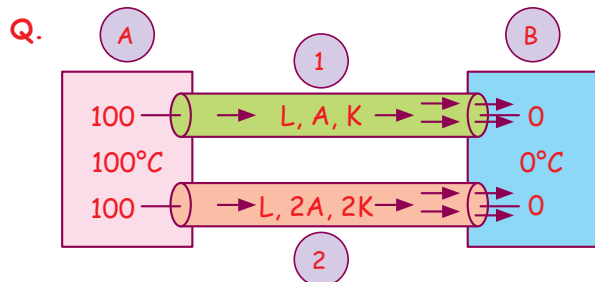
There is no accumulation of heat anywhere

$\left(\frac{dQ}{dt} \rightarrow \text{same}\right)$

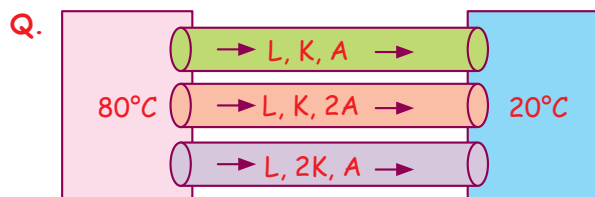
Sol. $\frac{dQ}{dt} = \frac{KA(80 - T)}{L} = \frac{2K \cdot 4A(T - 20)}{L}$

$$80 - T = 8T - 160$$

$$T = \frac{240}{9} = \frac{80}{3}$$



Sol. $\left(\frac{dQ}{dt}\right)_{\text{ऊपर}} = \frac{KA(100 - 0)}{L}$
 $\left(\frac{dQ}{dt}\right)_{\text{नीचे}} = \frac{(2K)(2A)(100 - 0)}{L}$
 $\left(\frac{dQ}{dt}\right)_{\text{net } A \rightarrow B} = (\text{ऊपर}) + (\text{नीचे})$
 $= \frac{KA(100 - 0)}{L} + \frac{2K \cdot 2A(100 - 0)}{L}$
 $= \frac{5KA}{L} \times 100$



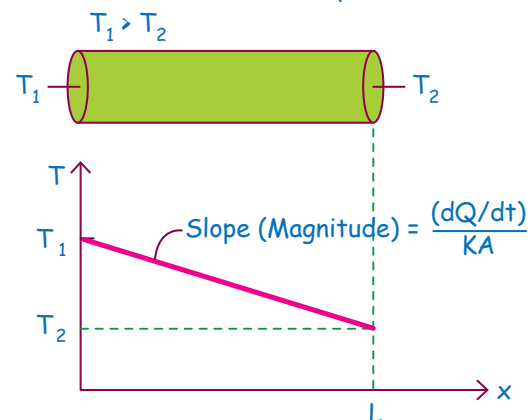
Sol. $\left(\frac{dQ}{dt}\right)_{\text{net}} = \frac{KA(80 - 20)}{L} + \frac{K(2A)(80 - 20)}{L} + \frac{2KA(80 - 20)}{L}$

Q. Suppose we have a rod whose temperature of one end is fixed at T_1 at $x = 0$ and another end is fixed at T_2 at $x = L$. If $T_1 > T_2$ then draw the graph between T vs x and analysis the situation.

Sol. $\therefore \frac{dQ}{dt} = -KA \frac{dT}{dx}$; $(T_1) \text{ fix at } x=0$ $(T_2) \text{ fix at } x=L$
 So, $\frac{dT}{dx} = -\frac{(dQ/dt)}{KA}$
 $\frac{dQ}{dt}$ is same for all cross-section area and if we plot graph between T vs x then $\frac{dT}{dx} = \text{slope}$
 So, $\frac{dT}{dx} = -\frac{(dQ/dt)}{KA} = \text{slope}$
 Slope $\propto \frac{1}{KA}$

If K and A is same everywhere $\Rightarrow$ slope will be constant graph will be straight line and after

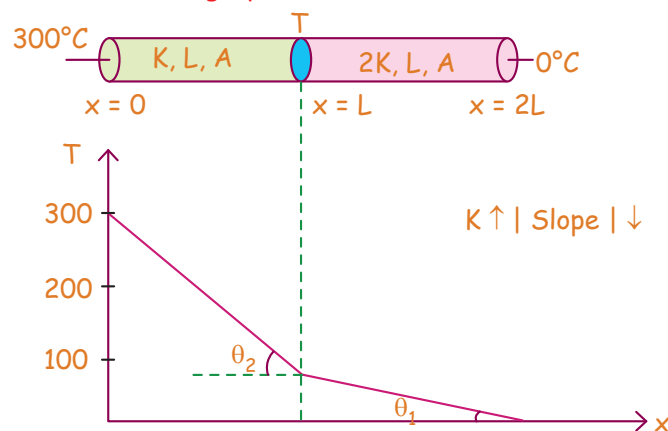
solving you can get $T(x) = \left\{ T_1 - \frac{(T_1 - T_2)}{L}(x) \right\}$



$$\frac{dQ}{dt} = -KA \frac{dT}{dx}$$

$$\frac{dT}{dx} = -\frac{(dQ/dt)}{KA} = \text{Slope (Constant)}$$

Q. Draw the graph between T vs x .



Sol. For temperature at junction solve

$$\frac{dQ}{dt} = \frac{KA}{L} (300 - T) = \frac{2KA}{L} (T - 0)$$

$$T = 100$$

इन सवालो को या ऐसे और मुश्किल सवालो को करने का एक मस्त तरीका है.... बताऊँ.... चलो रहने देते हैं



अब तक हमने देखा



$$\frac{dQ}{dt} = \frac{KA\Delta T}{l} = \frac{\Delta T}{l/KA}$$

$$\left(\frac{dQ}{dt} = \text{heat current}\right) = i \text{ (let)}$$

अब इन सवालों को करने का एक नया और मस्त concept पढ़ेंगे..... इसमें थोड़ा सा 10th class का current electricity use होगा समझने की कोशिश करना

$$\frac{dQ}{dt} = i = \frac{\Delta T}{l/KA} = \frac{\Delta T}{R_{th}} \Rightarrow i = \frac{\Delta V}{R} \text{ (याद करो 10th में पढ़ा था)}$$

जैसे CE में V (potential) होता था same ऐसे ही यहाँ temperature है, जैसे वहाँ ΔV होता था यहाँ ΔT है वहाँ high potential से low potential electric current flow करता था तो यहाँ high temperature to low temperature heat (heat current) flow करती है means

Heat flow through a conducting rod	Current flow through a resistance
Heat current $H = \frac{dQ}{dt}$ = rate of heat flow	Electric current $i = \frac{dq}{dt}$ = rate of charge flow
$H = \frac{\Delta T}{R}$	$i = \frac{\Delta V}{R}$

$$V \longrightarrow T$$

$$\Delta V \longrightarrow \Delta T$$

$$i \longrightarrow \frac{dQ}{dt}$$

$$i = \frac{\Delta V}{R} \longrightarrow \frac{dQ}{dt} = \frac{\Delta T}{l/KA} = \frac{\Delta T}{R_{th}}$$

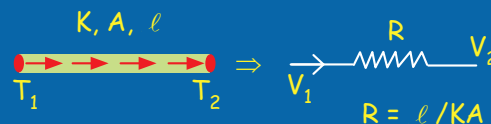
$$R = \rho \frac{l}{A} = \frac{l}{\sigma A} \longrightarrow R_{th} = \frac{l}{KA}$$

σ = electrical conductivity

K = thermal conductivity

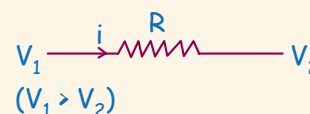
SKC

अरे समझ रहे हो ना..... मुद्दे की बात ये है की conducting rod को हम ऐसे resistance से replace कर सकते है जिसके resistance की value l/KA होगी



SKC बोले तो rod हटाओ तार बनाओ।

10th class revision

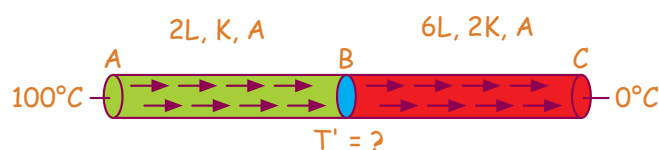


$$i = \frac{V_1 - V_2}{R} = \frac{\Delta V}{R} = i$$

$$\star 100^\circ\text{C} \xrightarrow{I, K, A} 0^\circ\text{C} \equiv \xrightarrow{i} \xrightarrow{R = l/KA} 100\text{ V} \xrightarrow{} 0\text{ V}$$

$$i = \frac{100 - 0}{R} = \frac{100}{l/KA}$$

Q. Find the heat current and temp T' at junction



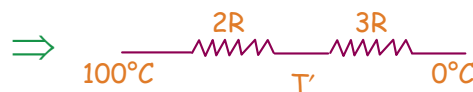
Sol. Method-I:

$$\frac{dQ}{dt} = \frac{KA(100 - T')}{2L} = \frac{2KA(T' - 0)}{6L}$$

Solve and get

Method-II: (SKC)

Rod हटाओ तार बनाओ Let $R = l/KA$



$$i = \frac{100}{5R} = \frac{20}{R} = \frac{20}{\frac{l}{KA}}$$

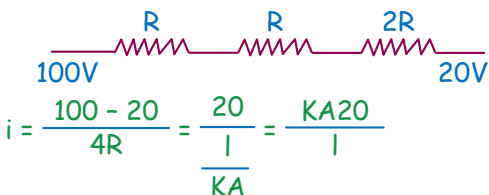
$$T' = 60^\circ\text{C}$$

Q. Find heat current.

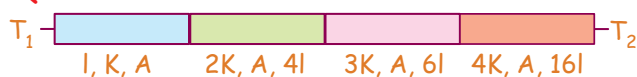


$$\frac{d\theta}{dt} = i = ?$$

Sol. $R = \frac{l}{KA}$ (let) so, above problem is analogous to



Q. Find the heat current.



Sol. $R = \frac{l}{KA}$ (let) so, above problem is analogous to

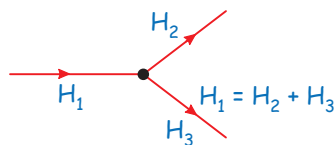
$$\frac{dQ}{dt} = i = \frac{T_1 - T_2}{R_{eq}}$$

$$= \frac{T_1 - T_2}{R + 2R + 2R + 4R} = \frac{T_1 - T_2}{9 \left(\frac{l}{KA} \right)}$$

• where $R = \frac{l}{KA}$

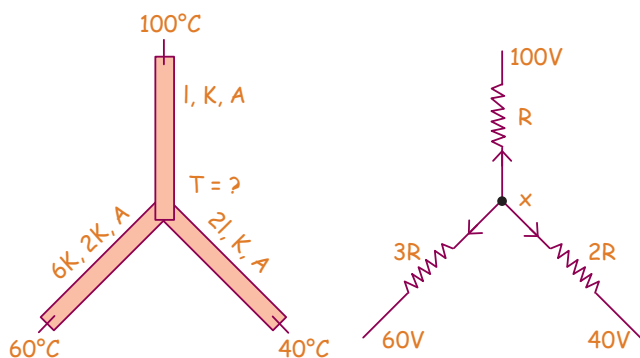
JUNCTION LAW

According to the Junction law the sum of all the heat current directed towards a point is equal to the sum of all the heat currents directed away from the points.



Similarly in current electricity we will study at the junction $i_1 = i_2 + i_3$

Q. Find temperature at the junction

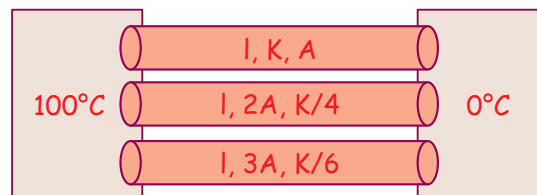


Heat Transfer

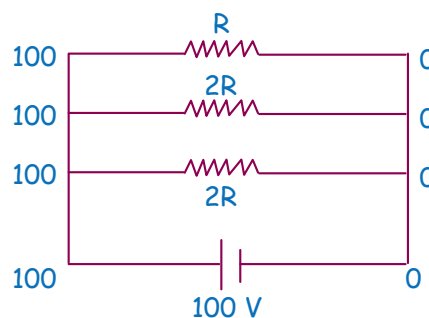
Sol. $i_1 + i_2 + i_3 = 0$ (Kirchoff's Junction Rule)

$$\frac{x - 100}{R} + \frac{x - 60}{3R} + \frac{x - 40}{2R} = 0 \text{ form this you can evaluate } x \text{ (ie. temp. at Junction)}$$

Q. Find net rate of heat flow.



Sol.



$$R_1 = \frac{l}{KA} = R \text{ (let)}$$

$$R_2 = \frac{l}{\frac{K}{4} 2A} = \frac{2l}{KA} = 2R$$

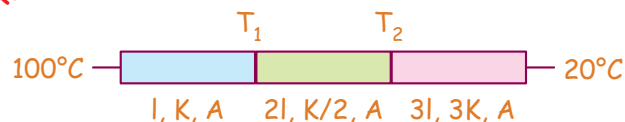
$$R_3 = \frac{l}{\frac{K}{6} 3A} = 2R$$

$$\frac{1}{R_{eq}} = \frac{1}{R} + \frac{1}{2R} + \frac{1}{2R}$$

$$R_{eq} = \frac{R}{2}$$

$$i = \frac{\Delta V}{R_{eq}} = \frac{100 - 0}{\frac{R}{2}}$$

Q.



Find T_1 , T_2 and $\frac{d\theta}{dt}$.



Sol.

$$i = \frac{80}{6R} = \frac{80KA}{6l}$$

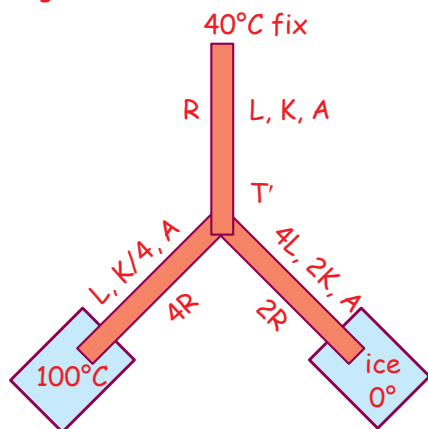
$$i = \frac{80}{6R} = \frac{100 - V_1}{R}$$

$$V_1 = 100 - \frac{80}{6} = \frac{520}{6}$$

$$i = \frac{80}{6R} = \frac{100 - V_2}{5R}$$

$$V_2 = 100 - \frac{400}{6} \Rightarrow V_2 = \frac{200}{6}$$

Q. Find temperature at junction and rate of melting of ice.



Sol.
$$\frac{T' - 40}{R} + \frac{T' - 100}{4R} + \frac{T' - 0}{2R} = 0$$

$$4(T' - 40) + (T' - 100) + 2T' = 0$$

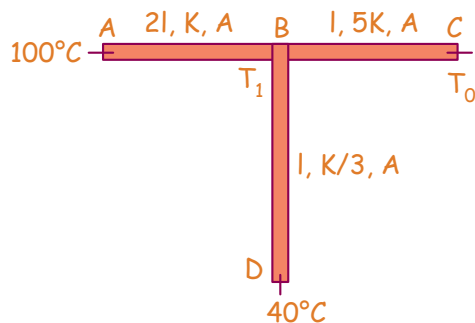
$$7T' = 260$$

$$T' = \frac{260}{7}^\circ\text{C}$$

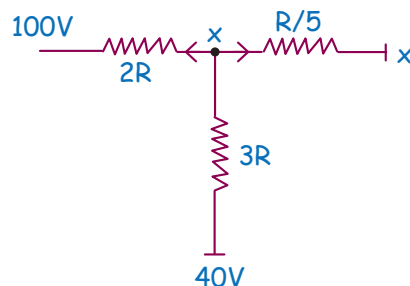
Rate of melting of ice =

$$\left(\frac{d\theta}{dt}\right)_{\text{ice}} = \frac{2KA(T' - 0)}{4L} = L_f \left(\frac{dm}{dt}\right)$$

Q. If heat current in rod BC is zero find $T_C = :$



Sol. Voltage - current equivalent diagram.

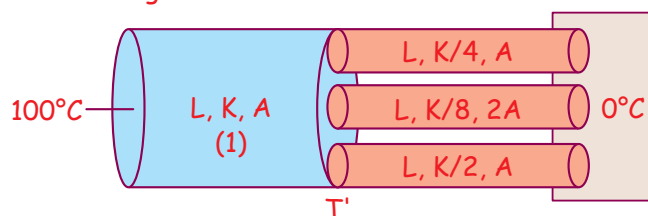


$$\frac{x - 100}{2R} + \frac{x - 40}{3R} + 0 = 0$$

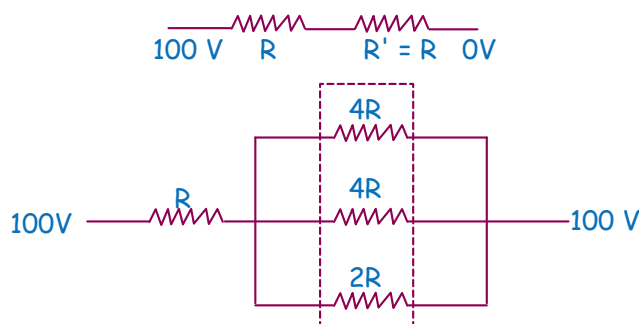
$$3x - 300 + 2x - 80 = 0$$

$$x = 76 \quad 76^\circ\text{C} \quad x = T_B = T_C = 76^\circ\text{C}.$$

Q. Find T' and net rate of flow of heat $\left(\frac{dQ}{dt}\right)$ through rod '1'.



Sol. $R = \frac{L}{KA}$ (let) then the current - voltage equivalent:



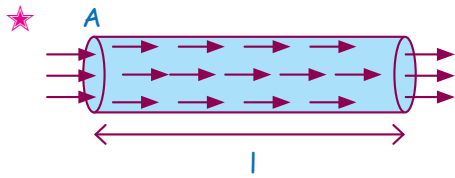
$$\frac{dQ}{dt} = i = \frac{100 - 0}{2R} = \frac{50}{R} = \frac{50}{\left(\frac{L}{KA}\right)}$$

$$T' = 50$$

अब हम अलग अलग body का thermal resistance निकालना सिखेंगे.... जो आपको 12th class के current electricity में as it is देखने के लिए मिलेंगे i will suggest अगर possible हो तो इन्हें याद कर लेना वैसे ज्यादा मुश्किल नहीं है।



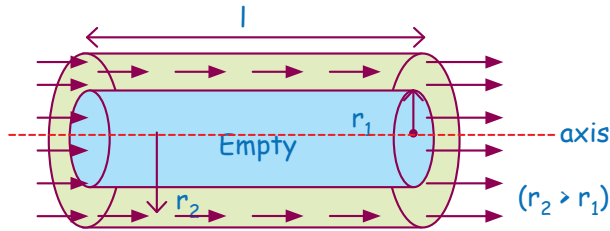
Thermal Resistance



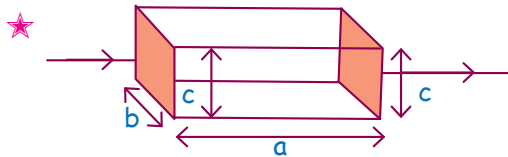
Solid cylinder (radius 'r')

$$R = \frac{l}{KA} = \frac{l}{K\pi r^2}$$

- ★ Hollow cylinder of inner radius r_1 and outer radius r_2 and heat is flowing along the axis.

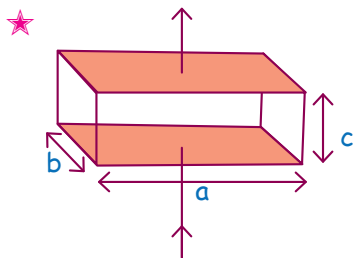


$$R = \frac{l}{KA} = \frac{l}{K\pi(R_2^2 - R_1^2)}$$



$$R_{eq} = \frac{l}{K(A)} = \frac{a}{Kbc}$$

cross section.



direction of heat flow.

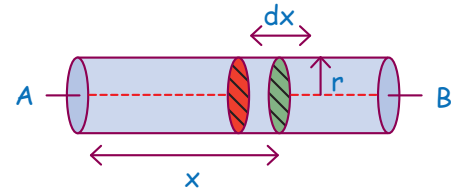
$$R_{eq} = \frac{c}{Kab}$$

- Q. Thermal conductivity varies with length (x) then find, R_{AB}

$$K = 2x + 3$$



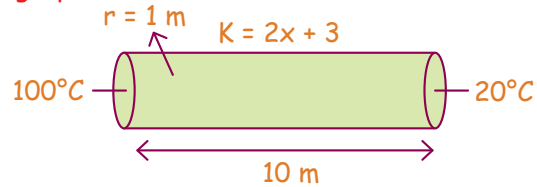
Sol.



$$R_{AB} = \int dR = \int_0^l \frac{dx}{(2x+3)\pi r^2} \quad \left(dR = \frac{dx}{KA} \right)$$

$$R_{AB} = \frac{1}{2\pi r^2} \ln(2x+3) \Big|_0^l = \frac{1}{2\pi r^2} \ln\left(\frac{2l+3}{3}\right)$$

- Q. Find rate of heat flow through the solid cylinder as shown in fig and R_{eq} if thermal conductivity of material varies as $K = 2x + 3$ length of the rod is 10 m and radius 1 m. Also draw the T vs x graph.



Sol. 1. $R_{eq} = ?$

$$\int dR = \int_0^{10} \frac{dx}{(2x+3)\pi r^2}$$

$$R_{eq} = \frac{1}{2\pi r^2} \ln\left(\frac{23}{3}\right)$$

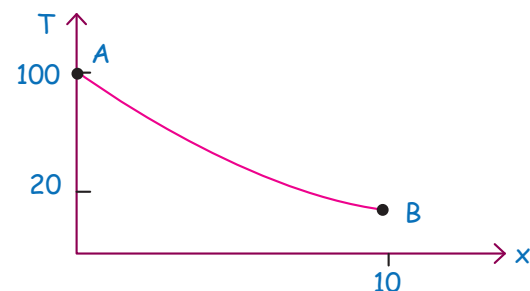
$$2. \left(\frac{d\theta}{dt} \right) = i = \frac{100 - 20}{R_{eq}}$$



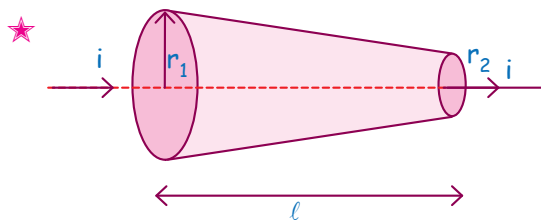
$$i = -KA \frac{dT}{dx}$$

$$\frac{dT}{dx} = \frac{-i}{KA} = \frac{-i}{(2x+3)A}$$

$x \uparrow$ | Slope | $\downarrow$



Very Important

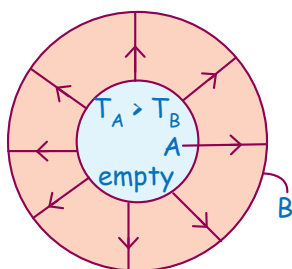


$$R_{eq} = \frac{\ell}{K\pi r_1 r_2}$$



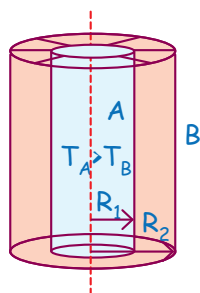
$$R = \frac{\ell}{K\pi r^2}$$

★ Hollow sphere (Heat flow Radially out)



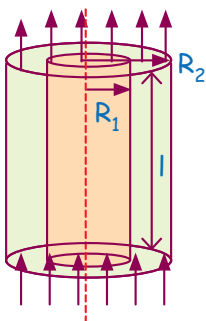
$$R_{eq} = \frac{R_2 - R_1}{K4\pi R_1 R_2}$$

★ Hollow cylinder (Heat flow radially out word)



$$R_{AB} = \frac{1}{K2\pi\ell} \ln \left(\frac{R_2}{R_1} \right)$$

★ Direction of heat flow



$$R = \frac{\ell}{K\pi(R_2^2 - R_1^2)}$$

C.E.

$$R = \frac{\ell}{KA} \equiv R = \frac{1}{\sigma} \frac{\ell}{A}$$

$$R = \rho \frac{\ell}{A}$$

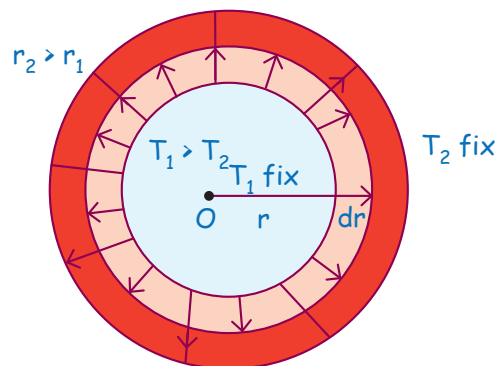
$$\frac{\ell}{K} \equiv \rho = \frac{1}{\sigma}$$

Hollow Sphere (thermal resistance derivation)

$$\int dR = \int_{R_1}^{R_2} \frac{dr}{K(4\pi r^2)} \quad \text{element will be a sphere of thickness 'dr' and radius 'r'}$$

$$R_{eq} = \frac{1}{4\pi K} \int_{R_1}^{R_2} \frac{1}{r^2} dr$$

$$R_{eq} = \frac{1}{4\pi K} \left(\frac{1}{R_1} - \frac{1}{R_2} \right) = \frac{1}{4\pi K} \left(\frac{R_2 - R_1}{R_1 R_2} \right)$$

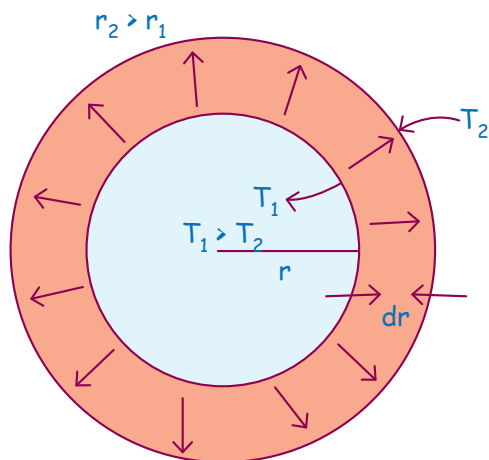


$$i = \frac{d\theta}{dt} = \frac{T_1 - T_2}{R_{eq}}$$

$$\int dR = \int_{r_1}^{r_2} \frac{dr}{K4\pi r^2}$$

$$R_{eq} = \frac{1}{4\pi K} \left(\frac{1}{r_1} - \frac{1}{r_2} \right)$$

$$\Rightarrow R_{eq} = \frac{1}{4\pi K} \left(\frac{r_2 - r_1}{r_1 r_2} \right)$$

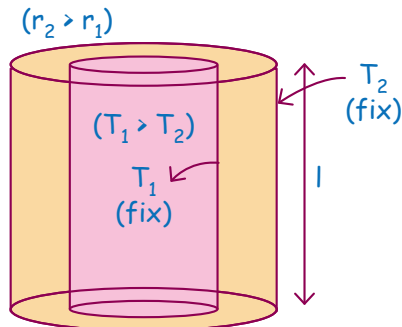


$$i = \frac{T_1 - T_2}{R_{eq}} = \frac{(T_1 - T_2)}{(r_2 - r_1)} 4\pi R r_1 r_2$$

Hollow Cylinder (thermal resistance derivation)

$$\int dR = \int_{r_1}^{r_2} \left(\frac{dr}{K(2\pi r l)} \right) = \frac{1}{2\pi k l} \int_{r_1}^{r_2} \frac{dr}{r}$$

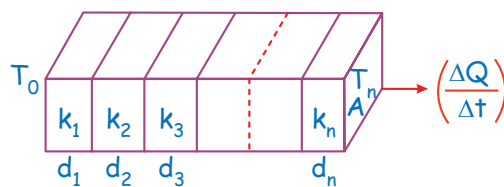
$$R_{eq} = \frac{1}{2\pi k l} \ln \left(\frac{r_2}{r_1} \right)$$



$$i = \frac{T_1 - T_2}{R_{eq}}$$

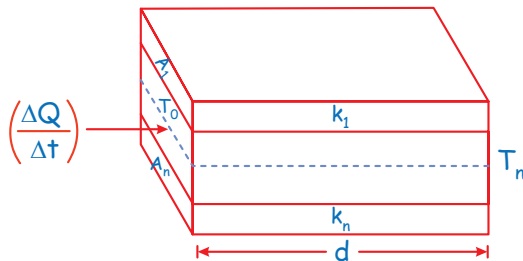
IMPORTANT RESULT

Layers in Series



$$\frac{\Delta Q}{\Delta t} = \frac{A(T_0 - T_n)}{\frac{d_1}{k_1} + \frac{d_2}{k_2} + \dots + \frac{d_n}{k_n}}$$

Layers in Parallel



$$\frac{\Delta Q}{\Delta t} = \frac{(T_0 - T_n)(k_1 A_1 + k_2 A_2 + \dots + k_n A_n)}{d}$$

Effective Thermal Conductivity in Series Combination

$$k_{eq} = \frac{d}{\frac{d_1}{k_1} + \frac{d_2}{k_2} + \frac{d_3}{k_3} + \dots + \frac{d_n}{k_n}}$$

Effective Thermal Conductivity in Parallel Combination: [Length will be Uniform]

$$k_{eq} = \frac{\frac{1}{A}}{\frac{1}{k_1 A_1} + \frac{1}{k_2 A_2} + \dots + \frac{1}{k_n A_n}}$$

Q. Two identical metal of thermal conductivities K_1 and K_2 respectively are connected in series. The effective thermal conductivity of the combination is:

Sol. Formula of thermal resistance, $R = \frac{l}{KA}$

$$R_{eq} = R_1 + R_2 = \frac{l}{K_1 A} + \frac{l}{K_2 A}$$

$$= \frac{l}{A} \left(\frac{1}{K_1} + \frac{1}{K_2} \right) = \frac{2l}{KA}$$

$$\Rightarrow K = \frac{2K_1 K_2}{K_1 + K_2}$$

l, K_1, A l, K_2, A

Q. Find 1. R_{eq} 2. K_{eq}

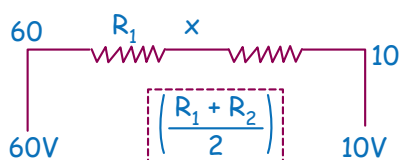
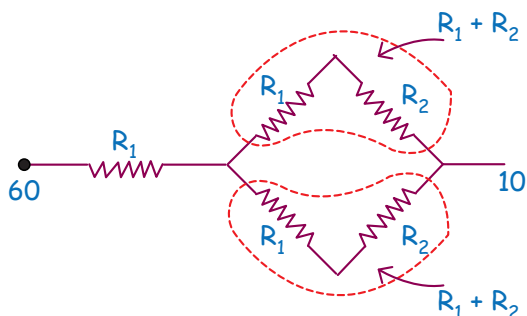
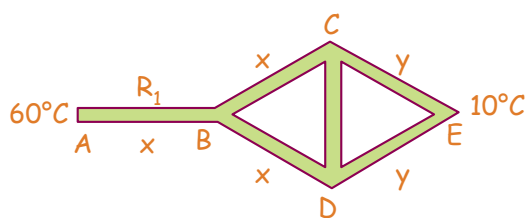
Sol. 1. $R_{eq} = \frac{l}{K_1 A} + \frac{l}{K_2 A}$ $R_{eq} = \frac{2l}{K_{eq} A}$

$$\frac{l}{K_1 A} + \frac{l}{K_2 A} = \frac{2l}{K_{eq} A} \quad \frac{K_1 + K_2}{K_1 K_2} = \frac{2}{K_{eq}}$$

$$K_{eq} = \frac{2K_1 K_2}{K_1 + K_2}$$

Q. Three rods of material x and three of material y are connected as shown in figure. All the rods are identical in length and cross sectional area. If the end A is maintained at 60°C and the junction E at 10°C , calculate the temperature of the junction B. The thermal conductivity of x is $800 \text{ W Wm}^{-1}\text{C}^{-1}$ and that of y is $400 \text{ Wm}^{-1}\text{C}^{-1}$.

$$R_1 = \frac{l}{K_1 A} \text{ (let)}$$



Sol. $\frac{x - 60}{R_1} + \frac{x - 10}{\left(\frac{R_1 + R_2}{2}\right)} = 0$

$$\frac{x - 60}{\frac{K_1 A}{L}} + \frac{2(x - 10)^2}{\frac{K_1 A}{L} + \frac{K_2 A}{L}} = 0$$

$$K_1 = 800 = 2K$$

$$K_2 = 400 = K$$

$$K_1(x - 60) + \frac{K_1 K_2}{(K_1 + K_2)} (2(x - 10)) = 0$$

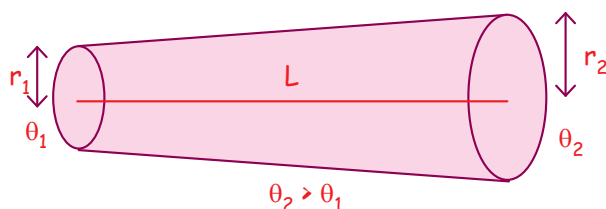
$$x - 60 + 2K_2 \frac{(x - 10)}{K_1 + K_2} = 0$$

$$x - 60 + \frac{2K(x - 10)}{3K} = 0$$

$$3x - 180 + 2x - 20 = 0$$

$$x = 40^\circ\text{C}$$

- Q. Find the rate of heat flow through a cross section of the rod shown in figure ($\theta_2 > \theta_1$). Thermal conductivity of the material of the rod is K.

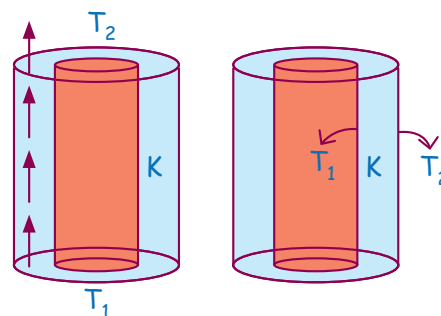


Sol. $i = \frac{\theta_2 - \theta_1}{R_{eq}} = \frac{(\theta_2 - \theta_1)}{L} K \pi r_1 r_2$

- Q. A hollow tube has a length l , inner radius R_1 and outer radius R_2 . The material has a thermal conductivity K . Find the heat flowing through the walls of the tube if (a) the flat ends are maintained at temperature T_1 and T_2 ($T_2 > T_1$) (b) the inside of the tube is maintained at temperature T_1 and the outside is maintained at T_2 .

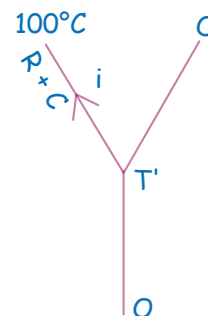
Sol.

$$\frac{l}{K \pi (r_2^2 - r_1^2)}$$



- Q. Three rods of Copper, Brass and steel are welded together to form a Y-shaped structure. Area of cross-section of each rod = 4 cm^2 . End of copper rod is maintained at 100°C where as ends of brass and steel are kept at 0°C . Lengths of the copper, brass and steel rods are 46, 13 and 12 cms respectively. The rods are thermally insulated from surrounding except at ends. Thermal conductivities of copper, brass and steel are 0.92, 0.26 and 0.12 CGS units respectively. Rate of heat flow through copper rod is:

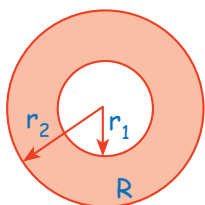
Sol. $T' = i = \frac{T' - 100}{R_1}$



- Q. Two thin metallic spherical shells of radii r_1 and r_2 ($r_1 < r_2$) are placed with their centres coinciding. A material of thermal conductivity K is filled in the space between the shells. The inner shell is maintained at temperature θ_1 and the outer shell at temperature θ_2 ($\theta_1 < \theta_2$). The rate at which heat flows radially through the material is:

Sol. Thermal resistance of spherical shell $= \frac{dr}{4\pi K r_1 r_2}$

$$\text{Thermal current } \frac{\Delta T}{R} = \frac{(\theta_2 - \theta_1)}{\left(\frac{r_2 - r_1}{4\pi K r_1 r_2} \right)} = \frac{4\pi K r_1 r_2 (\theta_2 - \theta_1)}{(r_2 - r_1)}$$



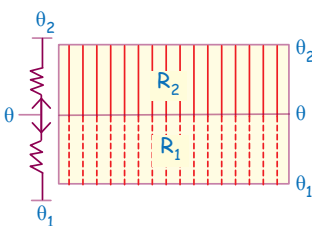
- Q. The temperature θ at the junction of two insulating sheets, having thermal resistances R_1 and R_2 as well as top and bottom temperatures θ_1 and θ_2 (as shown in figure) is given by:

Sol. $\frac{\theta - \theta_2}{R_1} + \frac{\theta - \theta_1}{R_2} = 0$

$$\frac{\theta}{R_2} - \frac{\theta_2}{R_2} + \frac{\theta}{R_1} - \frac{\theta_1}{R_1} = 0$$

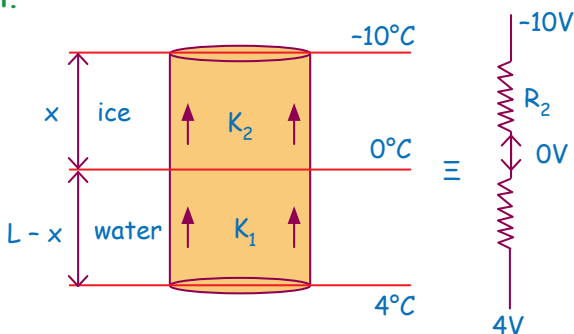
$$\theta \left(\frac{1}{R_1} + \frac{1}{R_2} \right) = \frac{Q_2}{R_2} + \frac{Q_1}{R_1}$$

$$\frac{\theta(R_1 + R_2)}{R_1 R_2} = \frac{\theta_2 R_1 + \theta_1 R_2}{R_1 R_2}$$



- Q. Consider a lake of depth L the temperature of base is constant at 4°C and atmosphere temperature is -10°C . Thermal conductivity of water is K_1 and for ice is K_2 . Find expression for maximum thickness of ice that can be formed theoretically.

Sol.

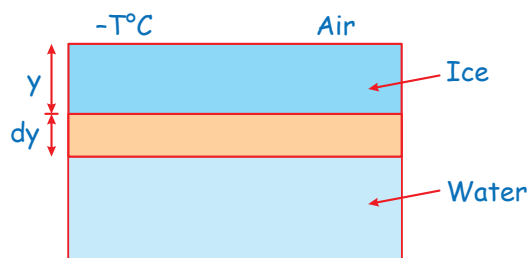


$$\frac{d\theta}{dt} = \frac{K_1 A (4 - 0)}{(L - x)} = \frac{K_2 A (0 - (-10))}{x}$$

GROWTH OF ICE ON LAKE

When the temperature of the air is less than 0°C , the cold air near the surface of the pond takes heat (latent) from the water which freezes it and a layer of ice is formed.

Consequently, the thickness of the ice layer keeps increasing with time. Let y be the thickness of ice layer in the lake at any instant t and atmospheric temperature be $-T^\circ\text{C}$. If the thickness is increased by dy in time dt , then the amount of heat flowing through the slab in time dt is given by $dQ_2 = mL = \rho(dt)A)L$ (see figure).



The temperature of water in contact with lower surface of ice will be zero. If A is the area of lake, heat escaping through ice in time dt ,

$$dQ_1 = \frac{KA[0 - (-T)]dt}{y}$$

As $dQ_1 = dQ_2$, hence, rate of growth of ice will be

$$\frac{dy}{dt} = \frac{kT}{\rho Ly}$$

So, the time taken by ice to grow to a thickness y is

$$t = \frac{\rho L}{kT} \int_0^y y dy = \frac{\rho L}{2kT} y^2$$

If the thickness is increased from y_1 to y_2 then time taken,

$$t = \frac{\rho L}{kT} \int_{y_1}^{y_2} y dy = \frac{\rho L}{2kT} (y_2^2 - y_1^2)$$

RADIATION

- ★ The process of the transfer of heat from one place to another place without heating the intervening medium is called radiation.
- ★ All objects emit radiations at above absolute zero temp ($T > 0 \text{ KE}$) and all objects absorb some of the radiation that falls on them from other objects.